

Inference of Numerical Competence from Speed and Accuracy

Anonymous CogSci submission

Abstract

People routinely infer others' competence under uncertainty, often relying on cues such as task's difficulty and past accuracy. An emerging body of research suggests that people approximate Bayesian inference when doing so. We extend these results by testing whether people can infer other's numerical ability in a way consistent with a rational Bayesian model. In Study 1, we found that participants accurately predicted math performance of another individual from information about past accuracy. Computational modelling shows that participants' inferences are better described by Bayesian processes than by plausible heuristics. Study 2 introduce a modified paradigm, where participants are told about both the past accuracy and speed of solving a problem. We found that, although participant are quite accurate in their predictions, they do not seem to take into account information about speed.

Keywords: Computational Modeling; Social Cognition; Competence; Knowledge Attribution; Information Search

Introduction

Humans routinely make decisions that depend on judgments of others' competence: whom to ask for advice, whom to hire, or whom to cooperate with (Cuddy et al., 2011; Harris et al., 2018; Xiang et al., 2023). Yet competence is not directly observable. Observers must infer it from sparse and noisy cues, and even apparently informative outcomes can be ambiguous (Jones, 1989). Consider choosing between two teachers after observing a single performance each: one solves an easy problem, the other fails a hard one. Despite the contrast between success and failure, neither observation is very diagnostic, because easy successes and hard failures are both common.

A growing body of work suggests that people nonetheless make competence inferences that approximate rational probabilistic reasoning (Aboody et al., 2025; Baker et al., 2017; Davis et al., 2025; Jansen et al., 2021; Jara-Ettinger & Gweon, 2017; Xiang et al., 2026). In particular, Mercier et al. (2025) used a naturalistic trivia paradigm to show that participants infer what others know from minimal evidence (a single success or failure), by combining the observed outcome with prior expectations about item difficulty and the average competence of the population in a domain. In this framework, competence judgments integrate (i) the diagnosticity of the observed performance—success on a hard item is more informative than success on an easy one—and (ii) prior beliefs about how competence varies across individuals.

However, previous research primarily concern knowledge-ability, a facet of competence (Aboody et al., 2025; Dubourg et al., 2025; Mercier et al., 2025). This leaves open a key question about generality: do people apply similar inference principles to skill-based competence? Many everyday judgments involve problem solving rather than the static retrieval of stored facts (e.g., mental arithmetic, programming). Here we test whether the same inferential logic extends to numerical competence, using arithmetic problems that vary in difficulty under a strict time limit.

Our contribution is twofold. First, in Study 1 we adapt the trivia paradigm of Mercier et al. (2025) and Dubourg et al. (2025) to arithmetic and test whether participants' judgments about another person's numerical ability are consistent with a Bayesian model that links latent competence to success probabilities across item difficulties. We also compare this normative account to heuristic baselines previously considered in the trivia domain (Mercier et al., 2025).

Second, in Study 2 we test whether observers additionally use speed as a cue to competence when accuracy information is held constant. While speed is often cited as a marker of arithmetic fluency (Cheng et al., 2021; Roy et al., 2025), it is theoretically ambiguous: faster responses may signal efficiency, but they can also reflect lower response caution, whereas for complex reasoning tasks, competent individuals often take longer to verify their answers (Berke et al., 2023; Goldhammer & Klein Entink, 2011; Hedge et al., 2018; Richardson & Keil, 2022). We therefore ask whether people's competence judgments track the true diagnostic value of response time in our task.

Computational Model

Inferring competence from performance requires accounting for uncertainty: people can guess, slip, or make occasional errors even when skilled (Jones, 1989). We formalize competence inference using the Bayesian framework developed in Mercier et al. (2025) and closely related to item-level models of ability (Jansen et al., 2021). Observers are assumed to represent a latent competence parameter θ for an individual and a known item difficulty parameter β for each task. After observing whether an individual succeeds ($S = 1$) or fails ($S = 0$) on an item of difficulty β , observers update beliefs about θ via Bayes' rule:

$$p(\theta \mid \beta, S) \propto p(S \mid \beta, \theta) p(\theta). \quad (1)$$

The likelihood $p(S \mid \beta, \theta)$ links competence and difficulty through a psychometric function (an IRT-style mapping), allowing outcomes to be stochastic rather than deterministic. Given the posterior over θ , the observer can then predict performance on new items by marginalizing over uncertainty in competence. The full specification of the likelihood and the prediction rule, along with model comparisons against heuristic baselines, is provided in the Methods for Study 1.

Study 1

In this study, we use a paradigm similar to Mercier et al. (2025) to investigate the inferences people make about individuals who provide either a correct or incorrect answer to a math question. We predicted that participants' behavior be best captured by the optimal Bayesian Model. More precisely, we put forward the following hypotheses¹:

H1. For different questions, participants can accurately assess the rarity of reaching an accurate answer within the time-limit.

H2. On average, participants' predictions of the probabilities of success on different questions, given the observation of a failure/success on a particular question, will be significantly correlated with true conditional probabilities.

H3. The predictions of the Main Model are positively correlated with participants' average predictions.

H4. When fitted to the aggregated dataset, the Main Model better captures participants' behavior than heuristic models.

Methods

Participants 300 U.S. participants were recruited via Prolific. We applied pre-registered exclusions for inattentiveness/LLM use ($N = 2$), suspected use of external calculation aids ($N = 35$), and manipulation noncompliance (changing the pre-ticked success/failure response; $N = 39$), leaving a sample of 224 (120 women, 104 men, $M_{Age} = 43.4$, $SD_{Age} = 14.8$)

Procedure This experiment replicates the paradigm of Mercier et al. (2025), but with different stimuli and a time constraint. Participants were asked to complete two phases. In the judgement phase, they were asked to evaluate five virtual agents on a set of fifteen mathematical questions. Participants were told that in order to help them, they will have access to the performance of the virtual agent on one of the questions. For each trial, participants were assigned to a Success/Failure condition: each virtual agent correctly ("Success" condition) or incorrectly ("Failure" condition) answered the question under 30 seconds. The specific question shown was randomly drawn from the stock of 15 questions. Only the question, and whether the virtual agent got it right or not, was provided to the participants, but not the specific answer. We call this item the "observed question."

¹The procedures, data collection, analysis plan and the models were pre-registered for Study 1 and 2: https://osf.io/aecyr/?view_only=2b8fc0d1c99e450f8b8534e2aab83a92. Data and code are openly available: https://osf.io/aecyr/files/github?view_only=2b8fc0d1c99e450f8b8534e2aab83a92

Participants were then asked to evaluate whether the virtual agent succeeded or failed to answer the other 14 questions from the quiz within the time limit. We call each of these items a "new question." Observed-new questions pairs are thus, for each observed question, the 14 predictions on new questions. They also had to estimate how many people, out of 100, correctly answered the observed question within 30 seconds.

In the second phase, participants had to complete the 15 maths questions, and one additional question designed to detect those using external tools. Questions' order was randomly determined. They were instructed that they had to answer each question under 30 seconds. At any time, once they had entered an answer, they could click Submit to proceed. At 15 seconds and 25 seconds time warnings appeared. If after 30 seconds they had not submitted an answer, the survey automatically proceeded to the next question and the timer restarted.

Materials The arithmetic stimuli were designed to establish a difficulty gradient following principles by Dehaene (2011). We manipulated five parameters to vary complexity: (1) answer magnitude (number of digits); (2) semantic roundness (operands ending in zeros vs. "sharp" non-zero integers); (3) operand count; (4) carry-over frequency (aggregate sums >10); and (5) placeholder tracking (alignment required for variable-length operands). For example, the problem $100,034,090 + 1,023$ represents a high-difficulty item as it requires placeholder tracking, mixed sharp/round operands, and a carry-over operation.

Computational Models We used the Bayesian/IRT framework from Mercier et al. (2025). Each question is assigned a difficulty score β by averaging judged difficulty ratings (divided by 100) and applying a logit transform to map scores from (0,1) to an unbounded scale.² Competence θ has a normal prior $\theta \sim \mathcal{N}(\mu, \sigma^2)$, and success is stochastic under a normal-ogive Rasch likelihood:

$$p(S = 1 \mid \beta, \theta) = \Phi\left(\frac{\theta - \beta}{\varepsilon}\right) \quad (2)$$

After observing $(\beta_{\text{obs}}, S_{\text{obs}})$, the posterior is proportional to $p(S_{\text{obs}} \mid \beta_{\text{obs}}, \theta) p(\theta)$. The predicted success on a new item is

$$P(S = 1 \mid \beta_{\text{new}}, \beta_{\text{obs}}, S_{\text{obs}}) = \int \Phi\left(\frac{\theta - \beta_{\text{new}}}{\varepsilon}\right) p(\theta \mid \beta_{\text{obs}}, S_{\text{obs}}) d\theta. \quad (3)$$

The Bayesian model has four free parameters: μ , σ^2 , ε , and a softmax temperature τ that maps probability estimates to binary responses. We compare this model to the same two heuristic baselines (Threshold, Anchor) described in Mercier et al. (2025).

²This transformation is inspired by the way difficulty is operationalized in frameworks like Item Response Rasch (1960). The logit transform is given by: $qlogis(p) = \log(p/(1 - p))$.

The Threshold heuristic assumes a deterministic rule: after a success, competence is at least the observed difficulty; after a failure, competence is at most that difficulty. It treats all values as equally likely, then predicts success on a new item by the posterior mass above its difficulty (with τ governing response noise). The Anchor heuristic collapses inference to a single point estimate: after success, competence is set a fixed distance Δ above the observed difficulty; after failure, Δ below. It then predicts success deterministically for items easier than that anchor (again with τ for response noise).

Results

Difficulty judgments (H1). Participants' perceived difficulty of each item closely tracked objective difficulty (i.e. the percentage of participants not being able to solve an item within the time limit in the questionnaire phase). A linear mixed-effects model predicting objective difficulty from perceived difficulty yielded a strong positive association ($\beta = 0.45$, $t(911.69) = 22.18$, $p < .001$; see Figure 1A).

Competence inferences from a single outcome (H2). Participants used the observed success/failure to make graded predictions about performance on other items. Averaged conditional predictions were strongly correlated with the true conditional probabilities computed from participants' own test performance ($\beta = 0.86$, $t(418) = 34.72$, $p < .001$, $R^2_{adj} = 0.74$; Figure 1B).

Model comparison (H3–H4). The Bayesian/IRT model reproduced participants' average conditional predictions across item pairs (Main Model: $\beta = 0.94$, $t(418) = 58.29$, $p < .001$, $R^2_{adj} = 0.89$), outperforming both heuristic baselines (Threshold: $R^2_{adj} = 0.39$; Anchor: $R^2_{adj} = 0.59$; Figure 2). Model selection using BIC favored the Bayesian model over Threshold and Anchor ($BIC_{Bayes} = 16063.72$, $BIC_{Threshold} = 19411.92$, $BIC_{Anchor} = 18259.96$), providing strong evidence for the Bayesian account.

Inferences among low-performing participants (H1m–H2m). To test whether accurate competence inference requires high arithmetic ability, we repeated H1–H2 in the bottom 30% of performers on the arithmetic test. Even in this subsample, perceived difficulty tracked objective difficulty (H1m: $\beta = 0.93$, $t(303) = 44.51$, $p < .001$), and average conditional predictions remained strongly correlated with true conditional probabilities (H2m: $\beta = 0.85$, $t(448) = 34.03$, $p < .001$, $R^2_{adj} = 0.72$).

Study 2

Study 2 tests whether observers incorporate response time as an additional cue to numerical competence, beyond accuracy. Response time is theoretically ambiguous: faster correct answers may signal fluency, but longer times can also reflect greater caution or verification (Berke et al., 2023; Goldhammer & Klein Entink, 2011; Hedge et al., 2018; Richardson

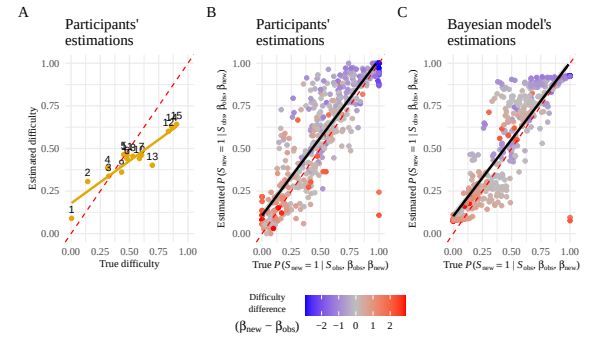


Figure 1: (A) Each question's estimated difficulty as a function of its true difficulty. Numbers refer to the question ID (see Materials). The red dashed line represents a perfectly accurate estimation. (B) Participants' estimated probability of answering a new question correctly, conditioned on the performance on the observed question, as a function of true conditional probabilities. (C) Our Bayesian model's estimated conditional probabilities as a function of true conditional probabilities. For (B) and (C), each data point corresponds to the average values for a pair of two questions: one for which the individual's performance is observed, and one for which the individual's performance has to be guessed. The red line represents a perfectly accurate prediction, while the dark line represents the fit of the linear model.

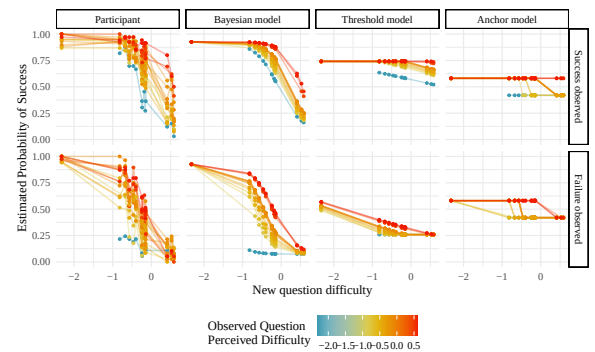


Figure 2: Model comparison. Each data point is a pair of two questions (one observed question and one evaluated question). The left column shows participants' average predictions, while the remaining columns show predictions from the Bayesian model, Threshold model, and Anchor model respectively. The top row shows trials where success was observed, and the bottom row shows trials where failure was observed. Lines connect predictions for the same observed question across different evaluated questions, colored by the observed question's perceived difficulty.

& Keil, 2022). To isolate the diagnostic value of speed, we introduced two observation conditions: in the Fast condition the target had 20 s to respond, whereas in the Slow condition the target had 40 s. The key idea is that observing a correct answer should be slightly more diagnostic of competence when it occurs under a tighter time limit. However, observing a failure should be less diagnostic of incompetence in the Fast condition compared to the same question in the Slow condition. All else equal, being in the Fast condition should lead to higher estimation of competence.

We pre-registered the following hypotheses:

H1. Participants accurately assess the difficulty of correctly answering the observed questions.

H2a. For a given pair of Quiz 1 and Quiz 2 questions, among participants who correctly answered the Quiz 2 question, those who answered under a 20-second time limit have a higher chance of answering the Quiz 1 question than those who answered under a 40-second time limit.

H2b. For a given pair of Quiz 1 and Quiz 2 questions, among participants who failed the Quiz 2 question, those who failed under a 20-second time limit have a higher chance of answering the Quiz 1 question than those who failed under a 40-second time limit.

H3. On average, participants' predictions of the probabilities of success on different evaluated questions, given success or failure within the 20-second or 40-second time limit on an observed question, are significantly correlated with true conditional probabilities.

H4a. For a given pair of Quiz 1 and Quiz 2 questions, participants' predictions of the probabilities of success on different evaluated questions will be higher for correctly answered observed questions under 20 seconds rather than under 40 seconds.

H4b. For a given pair of Quiz 1 and Quiz 2 questions, participants' predictions of the probabilities of success on different evaluated questions will be higher for incorrectly answered observed questions under 20 seconds rather than under 40 seconds.

Methods

Participants We recruited 400 U.S. participants via Prolific Academic. As in Study 1, we excluded participants who passed the cheat-test item ($N = 25$) or reported using calculation aids ($N = 18$); no one failed the attention check ($N = 0$). We additionally implemented bot-detection measures ($N = 7$). The final sample comprised $N = 354$ (224 women; $M_{Age} = 41.7$, $SD_{Age} = 12.7$).

Behavioral task As in Study 1, all participants completed both phases of the experiment - the Quiz Phase and the Judgement Phase. In Study 2, however, the Quiz Phase was completed first such that participants experienced answering the questions within different time limits before making judgements. During the Quiz Phase, participants took three arithmetic quizzes, each consisting of 5 questions. First, each participant took Quiz 1, a slow quiz where they had 40 seconds

to answer each question. Then each participant took Quiz 2 which could either be fast (20 seconds per question) or slow (40 seconds per question), with the time limit randomly assigned across participants. Finally, each participant took Quiz 3 which was assigned whichever time limit they did not have for Quiz 2. There was a distinct set of questions for each of the three quizzes (but constant across participants) and within each quiz, questions were presented in randomised order. At no point were participants shown the correct answers. In other words, we had two conditions randomised across participants: A "Fast Slow" Condition, where Quiz 1 allowed 40 seconds per question, Quiz 2 allowed 20 seconds per question (Fast) and Quiz 3 allowed 40 seconds per question (Slow), and a "Slow fast" Condition where Quiz 1 allowed 40 seconds per question, Quiz 2 allowed 40 seconds per question (Slow) and Quiz 3 allowed 20 seconds per question (Fast).

Quiz 1 and 2 performances were used as the 'true performance data', and compared to the judgements that people made in the Judgement Phase, where the same questions were used for the evaluated questions (Quiz 1) and the observed questions (Quiz 2). Quiz 3 was intended to ensure each participant experienced how difficult it is to answer questions within the different time limits. The cheat-test question was placed at the end of Quiz 1, but not included in any analysis not used as an evaluated question. As in Study 1, participants were instructed to only use mental calculation and told that their compensation did not depend on their performance. The Judgement Phase was almost identical to that in Study 1, except now participants also observed different time limits in which the virtual agent succeeded or failed on a given question (e.g. Participant id58 had a 40 second time limit to answer the following question, and we recorded that they got it right: $2646 + 3271$). Below the observed question, participants were presented the 5 questions from Quiz 1 in random order and asked to tick whether or not the virtual agent would succeed within the given limit. The perceived difficulty prompt for the observed question also integrated the time limit (e.g. Out of 100 participants, how many do you think got the question $2646 + 3271$ right within 40 seconds?). Participants assessed 5 virtual agents, with different observed question and time limit combinations for each, randomly assigned and balanced across participants.

Results

We first tested if participants were able to assess the difficulty of correctly answering the observed questions. We collected judgement of difficulty on the quiz 2 questions only as quiz 1 were materials from the first study. The analysis is not subdivided by time condition: both in the judgement phase and in the questionnaire phase, participants were presented with each condition half the time. A linear mixed effect model with random slope and intercept for participants showed that participants' estimations correctly tracked the difficulty of answering each question ($H1: \beta = 0.39$, $t(313.06) = 16.85$, $p < .001$).

To determine if the time condition in Quiz 2 provided information about the probability of answering correctly a question of the Quiz 1, we created pairs of Quiz 1 and Quiz 2 questions (e.g. Q1.1-Q2.1, Q1.1-Q2.2, etc.) from participants' answers. Note that this mirrors the structure of the judgement phase: while H2a and H2b test if the time condition provide information (for correct and incorrect questions respectively), H4a and H4b test if participants rely on the time condition when making a judgement. We ran two logits mixed-effect models: one among participants who answered a given Quiz 2 question correctly and one among those who answered it incorrectly. In both models, the dependent variable was a binary variable coding if the question from Q1 was correctly answered and the independent variable was a factor coding if the Q2 question was solved (H2a) or failed (H2b) under the Fast or the Slow condition. We controlled for the specific pair with a random effect (therefore comparing for the same pair, the effect of solving/failing under the fast condition compared to the slow), the model also include random intercepts for participants. H2a was restricted to trials where the Q2 question was solved and H2b was restricted to trials where the Q2 question was failed. We found that for a given pair of Quiz 1 and Quiz 2 questions, among participants who correctly answered the Quiz 2 question, those who answered under a 20-second time limit have a higher chance of answering the Quiz 1 question than those who answered under a 40-second time limit (H2a: $b = 0.51$, $z = 2.15$, $p = 0.031$). Among participants who failed the Quiz 2 question, those who failed under a 20-second time limit also had a higher chance of answering the Quiz 1 question compared to those who failed under a 40-second time limit (H2b: $b = 0.71$, $z = 2.63$, $p = 0.009$). This indicates that, no matter the outcome on the Quiz 2 question, participants should attribute higher chance of success to participants in the Fast condition, although the effect is relatively small.

We replicated the analysis of study 1 and found that, on average, participants' predictions of the probabilities of success on different evaluated questions, given success or failure within the 20 second or 40 second time limit on an observed question, are significantly correlated with true conditional probabilities (H3: $\beta = 0.89$, $t(98) = 19.47$, $p < .001$, Adjusted $R^2 = 0.79$). As shown in Figure 3, participants' estimations closely tracked the true conditional probabilities across both time conditions.

Mirroring the analyses of H2a and H2b, we tested if participants took into account the information provided by the time condition when predicting success/failure on evaluated Quiz 1 questions. Similar models to H2a and H2b were estimated, but with participants' binary predictions as the dependent variable and the observed time condition as the independent variable (still controlling for the pair presented and restricting the analyses among success/failure observed). Results show that participants do not take into account the information provided by the time condition when observing a correct answer under 20 second compared to 40 second (H4a: $b = 0.17$, $z = 1.53$, $p = 0.125$) or when observing an incorrect answer (H4b: $b = 0.15$, $z = 1.62$, $p = 0.104$).

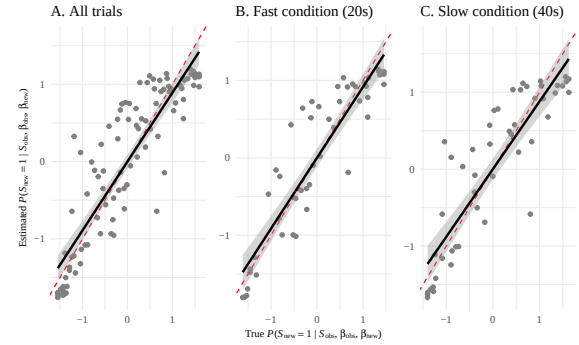


Figure 3: Estimated Conditional Probability versus True Conditional Probability in Study 2. Each data point is a pair of two questions. The red line represents the optimal prediction while the dark line represents the fit of the linear models. Panel A shows all trials combined, panel B shows trials in the Fast condition (20-second time limit), and panel C shows trials in the Slow condition (40-second time limit). Points are colored by the difficulty difference between the evaluated and observed questions.

We further tested if our results for H1 and H3 were robust when testing predictions of participants with minimal arithmetic ability (defined as the bottom 30% score in the questionnaire). Participants with minimal arithmetic ability can both accurately predict the difficulty of each question (H1m: $\beta = 0.32$, $t(88.04) = 8.16$, $p < .001$) and predict the conditional probabilities of success on evaluated questions (H3m: $\beta = 0.88$, $t(98) = 18.73$, $p < .001$, Adjusted $R^2 = 0.78$).

Moreover, participants' predictions of the conditional probabilities of success were still correlated with the true probability when analyzed at the trial level (H3i: $b = 1.95$, $z = 46.19$, $p < .001$).

General discussion and conclusion

The capacity to infer the competence and knowledge of others is a core component of social cognition. Our main focus was to test whether inferences about competence are consistent with normative Bayesian reasoning but our paradigm also allows us to test whether participants make judgments that approximate objective conditional probabilities well. Study 1 replicates previous

show that people can predict with a high degree of accuracy the conditional probability that a person knows the answer to a trivia question from a minimal amount of information—whether or not they know one other piece of information. This confirms that even a single observation can drive major updates in competence assessment, depending on the diagnosticity of the observed performance (e.g., success on a very difficult item). These studies also provide evidence that inferences of competence are well approximated by a Bayesian model which optimally integrates novel information with prior expectations. By contrast, plausible heuristics, corresponding

to lesioned versions of the Bayesian model, do not account as well for most participants' behavior. These two findings—accurate judgments, and reliance on normative inferences—are relatively independent, as applying non-normative heuristics sometimes results in accurate judgments, while normative inferences can produce biased answers when the generative model or the parameters are not well calibrated.

Our research opens new directions in understanding the fundamental mechanisms of social cognition. Building on prior literature establishing the importance of past accuracy and difficulty cues, the present research offers a systematic, computational account of how people utilize this information. We show that in a naturalistic paradigm, reliable cues, even if they are minimal, allow participants to form very accurate estimations of competence, in line with Bayesian principles. Future research could address the question of how participants weigh and integrate multiple cues of competence of varying diagnosticity.

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